

V4 RL Fine-tuning — Complete Analysis

Training Summary

Metric	Value
Episodes	500 (full run, no early stop)
Total generated	16,000 molecules
Final reward	0.544
Runtime	24.6 min (CPU)
Top-50 selected	Diverse, Tanimoto-filtered

Training Curve Assessment

Phase	Episodes	Reward	Validity	pIC50
Warmup	0–50	0.12–0.39	75–91%	5.5–5.7
Learning	50–200	0.38–0.54	91–100%	5.5–5.7
Stable	200–500	0.40–0.58	91–100%	5.6–5.7

Validity climbed from 84% → 100% — the model learned to avoid invalid SMILES.
Reward doubled from 0.29 → 0.54 — multi-objective optimization working.
Batch-average pIC50 stayed ~5.6 — but the *tail distribution* produced candidates up to 7.1.

Top-50 Candidates — Statistical Profile

Property	Min	Mean	Max	Drug Target
pIC50 (XGB)	6.67	6.82	7.11	≥ 6.0 ✓
QED	0.32	0.65	0.90	≥ 0.5 ✓
SA Score	1.86	2.61	4.28	≤ 6.0 ✓
MW	243	338	469	≤ 500 ✓
LogP	0.77	2.80	4.73	≤ 5.0 ✓
TPSA	35.0	67.7	119.9	≤ 140 ✓
Lipinski	—	100% pass	—	100% ✓

[!IMPORTANT]

All 50 candidates pass Lipinski's Rule of Five. 100% compliance.

Top-10 Medicinal Chemistry Deep Dive

#1 — C[C@@](C0)(Nc1ncnc2ccc(F)cc12)C1CC1

pIC50	QED	SA	MW	Verdict
7.11	0.89	3.1	261	★ Best overall

Scaffold: 6-fluoroquinazoline + cyclopropyl + chiral amine

Why it's good: Quinazoline is the exact scaffold of **erlotinib** and **gefitinib** (FDA-approved EGFR inhibitors). The fluorine at C6 enhances binding. Cyclopropyl adds metabolic stability. QED of 0.89 is exceptional.

#2 — Cc1cccc(Nc2ncnc(N[C@@]3(C)CCS(=O)(=O)C3)c2N)c1

pIC50	QED	SA	MW	Verdict
7.08	0.78	3.1	347	★ Strong

Scaffold: Diaminopyrimidine + sulfolane + toluidine

Why it's good: The 2,4-diaminopyrimidine mirrors **AZD9291 (osimertinib)** scaffold. Sulfolane improves solubility.

#5 — C0c1ccccc1Nc1ncnc(N2CCN(c3ccccc3)CC2)c1N

pIC50	QED	SA	MW	Verdict
7.01	0.71	2.1	376	★ Most synthesisable

Scaffold: Pyrimidine + piperazine + methoxyanilide

Why it's good: SA=2.1 means very easy to synthesise. The piperazine linker is a privileged kinase inhibitor motif. o-Methoxyaniline is a known EGFR-binding fragment.

#9 — c1ccc(Nc2ncnc(N3CCCC3)c2N)cn1

pIC50	QED	SA	MW	Verdict
6.89	0.87	2.2	256	★ Minimal & elegant

Scaffold: Diaminopyrimidine + pyrrolidine + pyridine

Why it's good: Only 256 Da — leaves massive room for lead optimization. QED of 0.87 is near-perfect. The compact structure is ideal for fragment-based drug design.

#11 — Nc1c(Nc2ccccc2)ncnc1N1CCSCC1

pIC50	QED	SA	MW	Verdict
6.87	0.90	2.3	287	★ Highest QED

Scaffold: Diaminopyrimidine + thiomorpholine + aniline

Why it's good: QED=0.90 is the highest in the entire set. Thiomorpholine adds unique SAR potential. Low MW (287) with good activity.

Scaffold Distribution — What the Model Learned

The model **converged on pyrimidine/quinazoline scaffolds** — exactly the right EGFR inhibitor chemotype:

Scaffold	Count in Top-50	Known EGFR Drugs
Pyrimidine (ncnc)	~30	Osimertinib, AZD3759
Quinazoline (ncnc+benzo)	~8	Erlotinib, Gefitinib
Thienopyrimidine	~5	TAK-285
Other heterocycles	~7	Various

[!TIP]

The dominance of pyrimidine/quinazoline scaffolds is a strong signal that the XGBoost oracle guided the generator toward EGFR-relevant chemical space.

Is This Production-Ready?

✓ What's production-ready

- **Generation pipeline end-to-end:** Pre-train → RL → score → select → output
- **All 50 candidates pass drug-likeness:** Lipinski, QED, SA, MW, LogP
- **Novel molecules:** Not in training set
- **Diverse set:** Tanimoto filter ensures scaffold variety
- **EGFR-relevant scaffolds:** Pyrimidine/quinazoline dominance

⚠ What needs more work before real deployment

1. **Quantum validation pending** — `quantum_pic50` is null for all candidates. When your QSVR model finishes training, we'll score these 50 candidates with the quantum oracle for a second opinion on binding affinity.
2. **Docking validation** — For a publication/production claim, these candidates should be docked against the EGFR crystal structure (PDB 1M17) using AutoDock Vina or Glide to confirm binding poses.
3. **pIC50 calibration** — The XGBoost model was trained on ChEMBL data. pIC50 of 7.1 means predicted IC50 $\approx$ 80 nM, which is plausible for lead compounds but should be validated experimentally.

📊 Practical verdict

For an AI/ML drug discovery project: This is a solid result. The pipeline generates novel, drug-like, synthesisable molecules that fall in the right scaffold classes for EGFR inhibition.

For actual drug development: These are *lead candidates* that would need computational validation (docking, MD simulation) and experimental testing (binding assays, cell viability).